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Problem 1

a) $Y \sim \text{Binom}(m, \pi)$

PMF of binomial distribution,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

here, $p = \pi$, $n = m$

$$P(Y = y) = \binom{m}{y} \pi^y (1-\pi)^{m-y}$$

$$\pi^y = e^{y \log \pi} \quad (\because \log \pi^y = y \log \pi \cdot 1)$$

$$\begin{aligned} (1-\pi)^{m-y} &= e^{(m-y) \log(1-\pi)} \quad (\text{similar to earlier}) \\ &= e^{(m \log(1-\pi) - y \log(1-\pi))} \end{aligned}$$

Now,

$$\begin{aligned} P(Y = y) &= \binom{m}{y} e^{y \log \pi} e^{m \log(1-\pi) - y \log(1-\pi)} \\ &= e^{\log \binom{m}{y}} e^{y \log \left(\frac{\pi}{1-\pi} \right)} e^{-m \log \left(\frac{1+\pi}{1-\pi} \right)} \\ &= \exp \left(y \log \left(\frac{\pi}{1-\pi} \right) - m \log \left(1 + \frac{\pi}{1-\pi} \right) + \log \binom{m}{y} \right) \end{aligned}$$

b) exponential family has PMF,

$$f_Y(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + \underbrace{c(y, \phi)}_{\text{---}} \right\}$$

So, eq (1) looks like exponential family pmf,

with $\theta = \log \left(\frac{\pi}{1-\pi} \right)$

$$v = n \dots$$

$$a(\phi) = 1$$

$$c(y, \phi) = \log\left(\frac{m}{y}\right)$$

$$b(\theta) = m \log\left(1 + \frac{\pi}{1-\pi}\right)$$

c)

The Expectation, $E(Y) = b'(\theta)$

$$= \frac{d}{d\theta} (b(\theta))$$

$$= \frac{d}{d\theta} \left(m \log\left(1 + \frac{\pi}{1-\pi}\right) \right)$$

$$= \frac{d}{d\theta} \left(m \log(1 + e^\theta) \right)$$

$$= m \cdot \frac{e^\theta}{1 + e^\theta}$$

$$\text{Variance, } \text{Var}(Y) = b''(\theta) a(\phi)$$

$$= \frac{d}{d\theta} \left(m \cdot \frac{e^\theta}{1 + e^\theta} \right) \cdot 1$$

$$= m \cdot \frac{e^\theta}{(1 + e^\theta)^2}$$

Canonical link function,

$$g(\mu) = \theta = \log\left(\frac{\pi}{1-\pi}\right)$$

d)

$$Y \sim \text{Pois}(\mu)$$

$$\text{PMF}, f(x, \lambda) = P(X=x) = \frac{\lambda^x \cdot e^{-\lambda}}{x!}$$

In our case,

$$P(Y=y) = \frac{\mu^y e^{-\mu}}{y!}$$

$$\mu^y = e^{y \cdot \log \mu}$$

(just like part a)

$$e^{-\mu} = e^{-\mu}$$

$$y! = e^{\log y!}$$

$$\text{So, } P(Y=y) = \frac{e^{y \log \mu} e^{-\mu}}{e^{\log y!}}$$

$$= \exp(y \log(\mu) - \mu - \log(y!))$$

→ (2)

e) exponential family PMF,

$$f_Y(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}$$

eq (2) clearly looks like this,

$$\text{sa } \left. \begin{aligned} \theta &= \log(\mu) \\ b(\theta) &= \mu = e^\theta \end{aligned} \right\} \text{ (ias } \theta = \log \mu)$$

$$a(\phi) = 1$$

$$c(y, \phi) = -\log(y)$$

f)

Expectation, $E(Y) = b'(\theta)$

$$= \frac{d}{d\theta} (\mu)$$

$$= \frac{d}{d\theta} (e^\theta) = e^\theta$$

Variance, $\text{Var}(Y) = b''(\theta) a(\phi)$

$$= \frac{d}{d\theta} (e^\theta) \cdot 1$$

$$= e^\theta$$

Canonical link function,

$$\eta(\mu) = \theta = \log \mu$$

HW6

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Problem 2

```
library(readr)
beetle_dat <- read_table("https://users.stat.ufl.edu/~winner/data/tribol_tox.dat",
col_names = FALSE)
```

```
##
## -- Column specification -----
## cols(
##   X1 = col_double(),
##   X2 = col_double(),
##   X3 = col_double(),
##   X4 = col_double()
## )
```

```
names(beetle_dat) <- c("pyr_conc", "pip_but_conc", "exp", "mort")
```

#Part a

```
bimodel <- glm(cbind(mort, exp - mort) ~ pyr_conc + pip_but_conc,
               data = beetle_dat, family = binomial)
summary(bimodel)
```

```
##
## Call:
## glm(formula = cbind(mort, exp - mort) ~ pyr_conc + pip_but_conc,
##      family = binomial, data = beetle_dat)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.77521    0.21048  -17.94   <2e-16 ***
## pyr_conc       4.53425    0.24630   18.41   <2e-16 ***
## pip_but_conc   0.28567    0.01706   16.74   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1038.05  on 12  degrees of freedom
## Residual deviance:  304.11  on 10  degrees of freedom
## AIC: 370.21
##
## Number of Fisher Scoring iterations: 4
```

#Part b The coefficient estimate for pyrethrin concentration is 4.53425. This means that, if we hold other predictors constant and increase pyrethrin concentration by one unit, the number of beetle mortalities will be multiplied by $\exp(4.53425)$, 93.153 times. This shows that pyrethrin concentration has very strong effect on the number of beetle mortalities.

#Part c Let's say the estimate for pip_but_conc is β , then the odds ratio would be

$$OddsRatio = e^{\beta(0.25-1.25)} = e^{-\beta}$$

From our summary of the model, $\beta = 0.28567$, so

$$OddsRatio = e^{-0.28567} = 0.7515$$

This means that, odds of death at piperonyl butoxide concentration of 0.25 are 0.7515 times that of when concentration is 1.25

#Part d Our model is, $\text{logit}(p) = \beta_0 + \beta_1 * \text{pyr_conc} + \beta_2 * \text{pip_but_conc}$ The estimated probability,

$$\hat{p} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot 0.5 + \beta_2 \cdot 0.3)}}$$

$$\hat{p} = \frac{1}{1 + e^{-(3.77521 + 4.53425 \cdot 0.5 + 0.28567 \cdot 0.3)}} = 0.194$$

#Part e

```
beta2 <- coef(bimodel)["pip_but_conc"]
odds_ratio <- exp(beta2 * (0.25 - 1.25))
cat("Odds Ratio for piperonyl butoxide concentration equal to 0.25 vs 1.25:", odds_ratio, "\n")
```

```
## Odds Ratio for piperonyl butoxide concentration equal to 0.25 vs 1.25: 0.7515142
```

```
beta0 <- coef(bimodel)["(Intercept)"]
beta1 <- coef(bimodel)["pyr_conc"]
logit <- beta0 + beta1 * 0.5 + beta2 * 0.3
p_hat <- 1 / (1 + exp(-logit))
cat("Estimated probability", p_hat, "\n")
```

```
## Estimated probability 0.1942878
```

```
datae <- data.frame(
  pyr_conc = c(0, 0.3),
  pip_but_conc = c(10, 5)
)

probe <- predict(bimodel, newdata = datae, type = "response")
cat("Predicted probabilities of death at given concentrations\n")
```

```
## Predicted probabilities of death at given concentrations
```

```
print(probe)
```

```
##           1           2
## 0.2852515 0.2715822
```

Problem 3 (a) False, For each distribution in the exponential family, link functions other than there is a canonical link function can also be used to construct a GLM. For example, we can use probit for binomial models. (b) True, For most exponential family distributions, the variance is a function of the mean. So, the variance changes as the mean changes. (c) False, The equation uses least squares to the log of the response variable, Poisson regression uses MLE. (d) True, The Poisson distribution can be used when something happens in a fixed time, the events here happen independently and at a steady average. So, in our case we can use Poisson distribution to count how many cats pass by.